

Lilja Kiiski

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EDUCATION

Purdue University, West Lafayette, IN

May 2029

B.S in Electrical Engineering, Honors | Minor in Mathematics

GPA: 3.86/4.00

Relevant coursework: Multivariate Calculus, Linear Algebra, C Programming, Intro to System-On Chip Design, Electrical Engineering Fundamentals

TECHNICAL / LEADERSHIP EXPERIENCE

Build@Scale Lab — Undergraduate Researcher

Incoming Aug 2026

West Lafayette, IN

- Incoming researcher focusing on design and fabrication of custom sensor interface electronics for instrumented manufacturing systems.

Purdue IEEE Autonomous Racing Kart Team — Electronics Lead

Aug 2025 - Present

West Lafayette, IN

- Designed the schematic for an 84V-to-19V/12V DC-DC converter PCB powering team's autonomous karts lidar system, working from system requirements and engineering drawings
- Simulated and validated core circuit in LTSpice under mentor guidance prior to physical layout and fabrication

Purdue Intro to System-On Chip Design Team — 4-Bit CSA Team Member

Jan - May 2026

West Lafayette, IN

- Collaborated within a team of 4 to design and build a physical 4-bit Carry-Save Adder summing three 4-bit numbers
- Developed a SystemVerilog testbench to conduct functional testing and report results across all 4,096 input combinations, verifying project functionality

Monta Vista Robotics Team — President, Computer Vision Lead

Oct 2021 - Jun 2025

Cupertino, CA

- Directed 150 member team through full design-build development of competitive FRC robot
- Led software division in development of team's first AprilTag computer vision localization and alignment system
- Worked with school district officials to approve funding for \$25M district-wide robotics facility

TECHNICAL PROJECTS

ENGR161 Mars Rover Autonomous LEGO Robot

Oct - Dec 2025

- Engineered autonomous robot to navigate terrain and deploy cargo upon magnetic field detection
- Programmed control logic for line-following navigation using Grove line sensors and IMU

Robot Computer Vision System

Oct 2022 - Apr 2025

- Built a Linux-based AprilTag detection pipeline enabling real-time robot self-localization on the game field
- Implemented alignment logic directing the robot to the nearest scoring location for game-piece deployment

Honeycomb Game Console

May - Jun 2025

- Engineered a miniature embedded game console with button/joystick input and a TFT LCD display
- Designed a custom honeycomb-inspired enclosure in Onshape, fabricated via 3D printing

PID Simulator

Jul - Aug 2022

- Developed a Java simulation tool modelling a robotic control system using a PID feedback loop
- Built configurable kP/kI/kD inputs with real-time visualization of simulated robot response

SKILLS

HDLs, EDA & Simulation Tools: SystemVerilog, KiCad, Altium (Learning), LTSpice, Onshape, Arduino

Hardware & Fabrication: Electronics (soldering, crimping, wiring), Basic machining (lathe, chop saw, table saw)

Software & Languages: C, C++, Java, Python, MATLAB, Linux, Bash/Shell, Git

Languages: English (Native), Finnish (Native), Spanish (Elementary)